In this work, we present a physics model capturing three key aspects of a two-phase partially molten mantle: (1) a degenerate Darcy–Stokes system describing the static mechanical behavior, (2) a nonlinear system governing the transport of chemical species and thermal energy, and (3) a eutectic phase relation characterizing melting processes. We develop a local mass conservative numerical framework to address this complicated multiphysics system.

For the static mechanics system, we introduce an $H(\mathrm{div})$-and an H^1- conforming mixed finite element spaces for Darcy and Stokes subproblems respectively. The mixed finite element spaces are constructed using a noval direct serendipity space tailored for quadrilateral mesh. A modified Bernardi–Raugel element is developed and analyzed. We provide detailed proofs of its stability and approximation properties. A new modified uzawa iterative method is implemented to solve the singular saddle point problem induced by this degenerate coupled system.

To address the nonlinear transport of chemical species and energy in the presence of porous media, we study and modify the weighted essentially non-oscillatory (WENO) method. We propose two simple alternatives to accelerate the evaluation of the smoothness indicator. Additionally, we develop a new multilevel WENO weighting scheme that is more versatile and flexible, allowing the combination of arbitrary stencils of different orders.

We advance the multiphysics system by incorporating a eutectic phase module that couples compositional and thermal information with porosity, thereby capturing the evolution of porosity and phase transitions.

The code is implemented in C/C++ along with PETSc, and is designed to be easily adapted for large-scale parallel execution on supercomputers, with the potential to support general unstructured meshes.